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Sum of two squares

Q: Which positive integers n can be expressed as

$$n = x^2 + y^2, \quad x, y \in \mathbb{Z} ?$$

n	$\square + \square$	n	$\square + \square$
1	$1^2 + 0^2$	13	$2^2 + 3^2$
2	$1^2 + 1^2$	14	X
3	X	15	X
4	$2^2 + 0^2$	16	$4^2 + 0^2$
5	$1^2 + 2^2$	17	$1^2 + 4^2$
6	X	18	$3^2 + 3^2$
7	X	19	X
8	$2^2 + 2^2$	20	$2^2 + 4^2$
9	$3^2 + 0^2$	21	X
10	$1^2 + 3^2$	22	X
11	X	23	X
12	X	24	X

Guess: For an odd prime p , $p = \square + \square$ if and only if $p \equiv 1 \pmod{4}$.

Recall that $-1 \in K p$ if and only if $p \equiv 1 \pmod{4}$.

Maybe the two phenomena are related?

Let

$$S = \{n \geq 1 : n = x^2 + y^2 \text{ for some } x, y \in \mathbb{Z}\}.$$

We saw $1, 2 \in S$ and $m^2 \in S$ for every integer $m \geq 1$.

Defn (Exponent) Let $n \geq 1$ be an integer and p a prime.

The exponent of p in n , denoted by $e_p(n)$, is the greatest

integer e s.t. $p^e | n$.

$$\text{Eg. } n = 1500 = 2^2 \cdot 3^1 \cdot 5^3.$$

$$e_2(n) = 2, \quad e_3(n) = 1, \quad e_5(n) = 3,$$

$$e_p(n) = 0, \quad p > 5.$$

Thm. Given an integer $n \geq 1$, $n \in S$ if and only if

for every $p | n$ with $p \equiv 3 \pmod{4}$, $e_p(n)$ is even.

Lem. Assume $n \in S$ can be written as

$$n = x^2 + y^2, \quad x, y \in \mathbb{Z}.$$

If $p | n$ and $p \equiv 3 \pmod{4}$, then $p | x$ and $p | y$.

Consequently, $p^2 | n$.

Pf. If $p | x$, then $p | x^2$. Since $p | n$, $p | y^2 = n - x^2$.

By Euclid, $p | y$. Moreover, $p^2 | x^2$ and $p^2 | y^2$. So

$$p^2 | n = x^2 + y^2.$$

Assume $p \nmid x$, $p \nmid y$. Look at

$$n = x^2 + y^2 \equiv 0 \pmod{p} \quad (*)$$

There is $x' \in \mathbb{Z}$ s.t. $xx' \equiv 1 \pmod{p}$, which is possible

because of $(p, x) = 1$. Multiplying $(*)$ by x'^2 :

$$\underbrace{(xx')^2}_{\equiv 1} + \underbrace{(yx')^2}_z \equiv 0 \pmod{p}$$

$$\Rightarrow 1 + z^2 \equiv 0 \pmod{p} \Rightarrow z^2 \equiv -1 \pmod{p}.$$

So $-1 \not\equiv p$. But this is impossible, as $p \equiv 3 \pmod{4}$. $\square$

Pf of the necessity part: Let $n \in S$ with

$$n = x^2 + y^2 \quad (**)$$

be minimal s.t. n has a prime factor $p \equiv 3 \pmod{4}$

with $e_p(n)$ odd. By Lem, $p \mid x$, $p \mid y$ and $p^2 \mid n$.

Dividing $(**)$ by p^2 gives

$$\underbrace{\frac{n}{p^2}}_{\in \mathbb{Z}} = \underbrace{\left(\frac{x}{p}\right)^2}_{\in \mathbb{Z}} + \underbrace{\left(\frac{y}{p}\right)^2}_{\in \mathbb{Z}} \in S,$$

where $\frac{n}{p^2} < n$ and $e_p\left(\frac{n}{p^2}\right) = e_p(n) - 2$ is odd. A

Contradiction. $\square$

Prop. S is multiplicatively closed: $m, n \in S \Rightarrow mn \in S$.

Pf. If $m = a^2 + b^2$ and $n = c^2 + d^2$, then

$$mn = (a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (bc + ad)^2 \in S.$$

$\underbrace{\hspace{1.5cm}}_{|a+bi|^2} \quad \underbrace{\hspace{1.5cm}}_{|c+di|^2} \quad \underbrace{\hspace{2.5cm}}_{|(a+bi)(c+di)|^2}$

$\underbrace{\hspace{2.5cm}}_{(ac-bd) + (bc+ad)i}$



Thm (Fermat - Euler) Every prime $p \equiv 1 \pmod{4}$ can be written as a sum of two squares.

Pf. Consider

$$x - ly, \quad x, y \in \{0, 1, \dots, \lfloor \sqrt{p} \rfloor\},$$

where we choose l later. There are

$$\underbrace{(\lfloor \sqrt{p} \rfloor + 1)}^{> \sqrt{p} - 1}^2 > (\sqrt{p})^2 = p$$

such pairs. Since there are only p residue classes mod p ,

there exist $(x_1, y_1) \neq (x_2, y_2)$ s.t.

$$x_1 - ly_1 \equiv x_2 - ly_2 \pmod{p}$$

$$\Rightarrow \underbrace{x_1 - x_2}_x \equiv l \underbrace{(y_1 - y_2)}_y \pmod{p} \quad (***)$$

At least one of x, y is nonzero, say x . Also,

$$0 < |x| \leq \lfloor \sqrt{p} \rfloor < \sqrt{p} \quad \text{and} \quad 0 \leq |y| \leq \lfloor \sqrt{p} \rfloor < \sqrt{p}.$$

Since $-1 \not\equiv 0 \pmod{p}$, we can find an $l \in \mathbb{Z}$ s.t. $l^2 \equiv -1 \pmod{p}$.

Squaring (***) leads to $x^2 \equiv l^2 y^2 \equiv -y^2 \pmod{p}$.

$$\text{So } x^2 + y^2 \equiv 0 \pmod{p}.$$

Thus $x^2 + y^2$ is a multiple of p . But

$$0 < x^2 + y^2 < (\sqrt{p})^2 + (\sqrt{p})^2 = 2p.$$

Hence $p = x^2 + y^2$.

